Team 3

Piaget's Constructivism

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Rethinking Science Education: Beyond Piagetian Constructivism Toward a Sociocultural Model of Teaching and Learning*

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Abstract

In the first part of the article I present an epistemological critique of forms of pedagogy founded on Piagetian constructivism. Despite the appeal of the notion that learners construct their understanding, I argue that constructivism is problematic because it ignores the subjectivity of the learner and the socially and historically situated nature of knowing; it denies the essentially collaborative and social nature of meaning making; and it privileges only one form of knowledge, namely, the technical rational. I then present a critique of active learning and student-centered forms of pedagogy. I argue that in our models of teaching we rely on too many unexamined assumptions from developmental psychology and we take for granted the problematic notion that children learn by doing. My central thesis is that constructivism is flawed because of its inability to come to grips with the essential issues of culture, power, and discourse in the classroom. In the concluding section of the article I present a preliminary account of a *sociocultural approach* to teaching and learning that takes seriously the notion that learning is situated in contexts, that students bring their own subjectivities and cultural perspectives to bear in constructing understanding, that issues of power exist in the classroom that need to be addressed, and that education into scientific ways of knowing requires understanding modes of classroom discourse and enabling students to negotiate these modes effectively so that they may master and critique scientific ways of knowing without, in the process, sacrificing their own personally and culturally constructed ways of knowing.

"Privileges only one form of knowledge, namely, the technical rational."

It exclusively favors technical-rational knowledge over other forms of understanding.

It only values one narrow type of thinking.

The author argues that (Piaget's) constructivism:

- It highly values cold, scientific logic and step-by-step problem-solving
- Ignored other valuable ways of knowing, such as emotional, creative, or cultural wisdom

Ignores individual and social context

- It overlooks the subjectivity of the learner as well as the socially and historically situated nature of how we acquire knowledge.
- It ignores who the student is and where they come from

- **Denies collaborative meaning-making:** It rejects or fails to recognize that making sense of the world is an essentially collaborative and social process.

- It assumes students can just figure things out on their own. In reality, we understand the world by talking, sharing, and working through ideas *together* with other people.

Relies on unexamined psychological assumptions

- Constructivism is strongly influenced by Piaget's developmental psychology.
- It assumes that learners naturally construct knowledge through cognitive development.
- According to O'Loughlin, these assumptions are often accepted without sufficient critical examination.
- Learning is treated mainly as an internal psychological process, while social, cultural, and historical influences receive less attention.

Takes *"Learning by Doing"* for granted

- It relies on the problematic notion that children naturally learn simply by doing.
- It separates “thinking” from “real-life”, learning is not just a physical activity.
- It's not truly "natural self-discovery"—it’s guided guessing.
- The author’s central thesis is that constructivism fails due to its **"inability to come to grips with the essential issues of culture, power, and discourse in the classroom."**
- Constructivism ignores unspoken rules, power dynamics (like who gets to speak and whose language is considered "correct"), and cultural barriers.

The Sims

https://www.youtube.com/watch?v=9KSFkcKBBMo





- Active construction of knowledge
- Prior knowledge
- Learning through experience
- Learning through problem solving
- Learning through exploration
- authenticity/ real world relevance
- Learner autonomy